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DRISHTI BIOTECH

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BIO MAGIC An organic based bio –stimulant with plant extracts-Treats chlorosis, increases growth and yield

REPORT ON REASEARCH, EXPERIMENTES & TESTS CONDUCTED | RAKESH JHA

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INTRODUCTION.

Chlorosis is a common occurrence all over the world (40% of world soils- Chen & Bank 1982, Shorrocks 1984). Groundnut is considered one of the most sensitive crops which suffer heavily from his bight sometimes resulting in 100% damage. The most pronounced effect is a drastic reduction in chlorophyll very much essential for biomass production (Lindsay, 1982, 1984). It has been reported that at all the stages, i.e., flower production, peg initiation, peg penetration, pod initiation, pod formation, an pod filling adequate supply of photosynthate is essential (Williams et al 1975, 1978 and Krishnashastri et al 1985). At the international level 60 institutions are engaged in researching cost effective solutions to this problem and even after decades of research iron deficiency remains an important plant nutrition problem (Chaney & Bell 197, Katyal & Sharma 1985).

MANAGEMENT

Two principal methods of treating iron deficiency are followed:

A. Soil Treatment : In soil treatment with iron containing fertilizers synthetic iron chelates are effective but because of the very high cost (in India Rs 2.5 Lakhs/ha) they are cost prohibitive. Application of ferrous sulphate does not give repeat performance and also large quantities are required.

B. Foliar Treatment : Foliar sprays of inorganic salts have been found to be of benefit but require frequent treatment. In the category of foliar spraying of 2 % ferrous sulphate and ferrous sulphate + Citric acid (5:1 of 10:1) are recommended but farmers have not fully adopted this method as on many occasions phyototoxic effect with stunting of plants have been observed.

Most frequently it is observed that in soils with iron deficiency, zinc and manganese deficiencies have also been noted to occur. BIO MAGIC takes care of these deficiencies have also been noted to occur. BIO MAGIC takes care of simultaneously suffer from nitrogen deficiency as nitrogen fixing enzymes also require iron. In Cyprus a 45% increase in yield of groundnut was obtained by foliar spray (Papastylianou 1990). In Thailand a 60% increase was recorded by foliar spray (Ratanrat et al 1988). There are several other reports where significant results have been achieved with appropriate foliar sprays. Dangule & Zende (1976) reported that foliar application of Miscnelf (Ms 10) in two sprays at 35 and 50 DAS increased groundnut yield by about 12% over the control. After considering all the above aspects, the manufactures tried to develop sequestered compounds having balanced proportion of micronutrients appropriately blended with microbial extract and chelating agent. Results of BIO MAGIC were very encouraging in pto and mini –plot experiments as well as in 22 field trials conducted in problem areas. Results indicated a 28% increase in yield in all there moisture regimes tired. According to current available literature on iron chlorosis, the general cocenus amongst scientists is that foliar spray of iron containing formulations is the most practical remedy having both the advantage of cost – effectiveness and avoiding the need for frequent applications. BIO MAGIC is the answer. It combines the advantages of being cheap, very concentrated and requiring few applications.

EFFECT OF BIO MAGIC ON PADDY AND CHILLY CROPS

Seeing the very good effect of BIO MAGIC in correcting chlorosis and improving health and final yield of groundnut, it was decided to test its efficacy on paddy and chilies. Paddy seedlings and Paddy showing symptoms of chlorosis totally recovered after just now spray applications of BIO MAGIC is the leaves turned dark green and the whole plant appeared healthier. The same results were obtained in trails with chillies.

REASEARCH DATA

Tabulated hereunder are the results of some of the data culled from field trails. It will be noted that every effort has been made to compare BIO MAGIC with other treatments and also the different types of application of BIO MAGIC as well as the application of BIO MAGIC in various concentrations TO determine the best concentration and method of application.

Table 1

RESULTS OF BIO MAGIC ON VARIOUS DEMONSTRATIONS ON FARMERS' FIELDS IN INDIA

DISTRICT	AGENCY	NUMBEROF DEMONSTRATIONS	INCREASE IN YIELD
Junagadh	Aga Khan Rural Support prog. Training & Welfare Scheme	29	18%
Bhavnagar	Government of Gujarat Training & Welfare Scheme	8	12%
Rajkot	Government of Gujarat Training & Welfare Scheme	25	12%
Kheda	Government of Gujarat Training & Welfare Scheme	14	15%

In fields where there was proper rainfall and adequate sunshine the increase in yield was above 25% .In on field in the Kheda districts there was an increase of 5o% in yield and Rajkot one field showed a 27% yield increase despite there being no rainfall.

Table 2

EFFECT ON BIO MAGIC ON VARIOUS PARAMETERS OF GROUNDNUT CULTIVATION

RESULTS	**WITH BIO MAGIC (TRIAL)**	**WITHOUT BIO MAGIC (CONTROL)**	**CHANGE (% Age)**
Number of roots per plant	36.60	28.15	+30.02%
Number of pods per plant	29.80	12.51	+138.21%
Average Weight of each pod	10.86	8.96	+21.21%
Weight of pods per Ha (kg)	3104	2591	+19.80%
Percentage of kernel	72.34	66.35	+9.03%
Percentage of oil	49.86	48.56	+2.88%
Percentage of protein	24.89	20.41	+21.95%
Percentage of carbohydrate	30.37	25.08	+21.09%
Production per hectare (kg)	1114.00	860.00	+29.53%
TOTAL	102.60	66.60	+54.05%

TABLE -3
CHLOROPHYLL CONTENT IN SAPOTA LEAVES TREATED WITH BIO MAGIC (Sapota Orchard Test Farm , Anand , Gujarat , India,1995)

Sample	**CONTROL GROUP**			**TEST GROUP**			**Change**
	Chlorophyll A	Chlorophyll B	Total Chlorophyll	Chlorophyll A	Chlorophyll B	Total Chlorophyll	%
Sample 1	2.6216	1.1612	3.9699	5.1877	1.8962	7.2879	+83.58
Sample 2	2.4696	1.3706	4.0410	7.3088	3.1841	11.0102	+172.46
Sample 3	2.9537	2.0805	5.3131	4.4474	2.1003	6.8768	+29.43
Sample 4	3.4790	1.6895	5.4031	9.1615	4.1023	13.922	+156.39
Sample 5	2.3951	1.5955	4.2088	5.2145	3.0457	8.6974	+66.79
Average			4.5926			9.5589	+66.79

(Chlorophyll Content –Mean of two replications in milligrams /gram/ fresh weight)
The above tests were conducted under the aegis of the Department of Biochemistry , College of Agriculture, Anand Campus , Gujarat Agricultural University , Gujarat India .

Table -4
EFFECT OF SOIL AMENDMENTS AND FOLIAR SPRAYS OF BIO MAGIC ON YIELD OF SUMMER GROUNDNUT

	TREATMENT	POD YIELD		POD	% INCREASE
		1992	1993		
1	BIO MAGIC (Soil)	1151	2745	1948	+5.01%
2	BIO MAGIC (Foliar)	1533	3139	2337	+25.96%
3	Ferrous Sulphate (Soil)	1007	2385	1696	-8.57%
4	Ferrous Sulphate +Ca (Foliar)	1226	2782	2004	+8.03%
5	25kg N/Ha	1408	2752	2080	+12.13%
6	Absolute Control	1082	2627	1855	0.00

Considering two years pooled results, foliar spray of BIO MAGIC was significantly superior to other treatments, whereas in individual years treatment differences were non- significant but when degree of freedom increased the difference came to be significant in pooled data. The data presented in this experiments is quite in agreement with the recommendation already adopted in joint Agresco (May 93) when 28.0% increase in yield was reported . Soil treatment with ferrous sulphate Of BIO MAGIC proved non effective.

Table 5
EFFECT ON GROWTH STIMULANTS ON THE YIELD OF GROUNDNUT

TREATMENT	POD YIELD(Kg/Ha)	% CHANGE OVER CONTROL
NAA @ 20ppm	2649	+6.0%
G.A.@ 20ppm	2367	-5.3 5
Boric Acid @ 0.2%	2642	+5.7%
AKTI M -3	3275	+31.0%
FeSO4+ Citric Acid (5:1)	2770	+10.2%
Control (No Spray)	2499	0.00
S. Em	162.6	
C.D. at 5%	489.7	
C.V %	12.03	

The above data indicate the superiority of BIO MAGIC -3 sprays at intervals of 15 days commencing from 30 DAS registering 31% increase in yield. This experiment also showed that conventional treatments of FeSo4+ Citric Acid spray was not as effective in increasing yield as BIO MAGIC , The data also occur with earlier recommendation adopted at Joint Agresco, May 1993, when a 28% increase in yield was recorded.

Considering the results ,i.e. three years in Agronomy Department (from where the recommendation emerged), two years collecting data from experiments undertaken based on suggestion from committee on chlorosis in groundnut regulators, BIO MAGIC has proved itself an effective agent in increasing the yield of groundnut by between 25 to 30%.

PLATE-I

This plate shows general layout of the experiment conducted to study the efficacy of BIO MAGIC on groundnut and paddy in areas where these crops are grown extensively. Soils were from various districts in India (Amreli, Panchmahals, Junagadh, Bhavnagar). Four tonnes of soil were brought from each district following standard procedure of collecting soil samples. The soil were thoroughly mixed and dividend into two equal portions of two tonnes each and dispensed in 2m x 1m 1m x 1m mini plots as shown in this photograph. The districts from which soil was collected have about 1,00,000 hq under groundnuts cultivation during the Kharif season (June to November). Crops suffer from mid to severe chlorosis and the recommended practice of 3 to 4 sprays of ferrous sulphate plus citric acid was not adopted by the farmers. BIO MAGIC managed to successfully control the chlorosis. The above photograph clearly shows the difference between treated and untreated plots and the effect of BIO MAGIC in controlling chlorosis.

PLATE II

The above photograph shows a close-up view of the mini –plots of Amreli soil, pH 7.75.caco3 31% well drained shallow soils suited for groundnut , cotton, millet ,sorghum, etc. During the monsoon season bunch-type groundnuts are grown. The soils being calcareous in nature calcium induced iron deficiencies occur resulting in chlorosis. Crops show chlorosis , sometimes within a week of germination , This photograph was taken 36 days after sowing (D.A.S.) and after one spray of BIO MAGIC (after 30 D.A.S). The plot towards the left was not treated and is clearly showing the effect of chlorosis.

PLATE III

This photograph above shows chloratic conditions prevalent in the soils Junagadh District Here, groundnut plantation covers an area of 600,000hq . and chlorosis is observed in 50 % of the area. This hampers overall productivity of the groundnut plantation. The plot on the left has received just one spray of BIO MAGIC 0.5% D.A.S. The difference is remarkable.

PLATE IV

This photograph shows yet another set of plots containing soil from the Junagadh District but with a higher clay content and comparatively lower calcium carbonate (16%). The plot is however , still severly contaminated by chlorosis. One spray of BIO MAGIC 0.5% at D.A.S showed considerable improvement.

PLATE V

In this photograph , the effects of chlorosis are highlighted. Marginal necrosis and stunted growth was observed . Death soon followed. In a plot of soil from Jundagadh District , 60 groundnut seeds were planted, but over time , hardly 50% of the seedlings survived. Those that did , had less than expected growth and yield. In severe cases , complete loss of yield have been reported.

PLATE VI

This photograph may seem similar to that in Plate V expect that the soil is from a different district, that of the Amreli District . As is obvious , the plants are stunted and there are more dead plants. In the plots treated with BIO MAGIC, a survival rate of between 95-100% was recorded resulting in a significant increase in yield.

PLATE VII

After being treated with only three sprays at 30 D.A.S and 60 D.A.S respectively, this photograph showing micro- plots of Amreli soil show obvious results. The plot on the right is rich with colour. Chlorophyll contents were on average 3mg/g of fresh weight compared to 1.75mg/g recorded in control. The increase in yield is about 9% (pods).

PLATE VIII

That is a close-up photograph of Plate VII. The cumulative effect of earlier treatment is evident on the lower leaves. Marked green spots on chloratic leaves were observed within three days of first being treated with BIO MAGIC.

PLATE IX

Here we have a close-up photograph of the effect of treatment on groundnut plants sown in Amreli soil. Notice that the lower leaves are darker green, the direct effect of two earlier.

PLATE X

This is a photograph of a plot not treated with AKTI-M3. Inter venal chlorosis is evident and some of the leaves are beginning to show marginal necrosis. In severe cases, this resulted in the death of the plants seen in Plate V and Plate VI. Note that the leaves are smaller in size than compared to those in the next plate, XI.

PLATE XI A

To study the increase in chlorophyll content, a technique was developed to treat half of the leaf with BIO MAGIC . This eliminated the possibility of differences arising out of samples drawn from different physiological stages of leaf development. The increase in chlorophyll content was noted from day two onwards after treatment with BIO MAGIC.

PLATE XI B

On average leaves treated with BIO MAGIC had a chlorophyll content of 3.1mg/g of fresh weight an increase of over 180% compared to the chlorophyll content of the control which was 1.1mg/g fresh weight. In the picture above the second leaf from the bud of the groundnut plant began to green within only 24 hours after being treated with BIO MAGIC.

PLATE XII

Paddy seedlings and upland paddy are not immune from chlorosis. In Gujarat state, paddy is grown in an area about 300,000 hq. and some of the cultivators suffered severely from chlorosis. The soil seen in this photograph was brought from the Panchmahal District. The left mini-plot is chloratic and almost all the seedlings have died. The right mini-plot however, received just one spray of BIO MAGIC 11 days after germination. Almost all the seedlings in this mini-plot survived. The difference is clear.

PLATE XIII

A mini-plot measuring 2m X 1m was sub- dividend into 24 micro-plots . Six different treatments were tired to alleviate chlorosis. As can be seen in this photograph, micro-plot treated with 0.5% BIO MAGIC survived. The survival of other micro-plots were however, negligible.

PLATE XIV

Citrus Sp. Suffers from chlorosis when grown in calcareous soils because they are highly sensitive to Irion and Zinc deficiency. This may lead to Die –back disease. Plants are stunted and fruit development retarded. In severe cases, plants die completely. The photograph above shows the alleviation of chlorosis after 4 sprays of 0.5% BIO MAGIC at intervals of 12 to 15 days.

PLATE XV

This photograph shows how quality assurance is monitored. Each batch of BIO MAGIC is treated on live plants with concentrations of 0.025% to 1% on half the leaf. Optimal chlorophyll content is observed in 0.5% concentration. The same technique

can be used for estimating chlorophyll periodically. No batch of BIO MAGIC will leave the production site until tested in this manner.

PLATE XVI

As mentioned earlier, chlorophyll is monitored using the 'half lea' technique where half the leave is treated and the other half kept as control. Leaf tissue weighing 0.1g is cut from the treated half as well as the untreated half of the some leaf and is incubated in D.M.S.O reagent for 1 hour. Chlorophyll a,b and total are measured by taking readings at appropriate wavelengths in the spectrophotometer.

PLATE XVII

After successful trials in potted and mini-plots, a full – scale field trail was conducted at the University Experimental Station. These trails were conducted in the calcareous soils of the Junagadh Districts (31% Calcium Carbonate, pH 7.85). The soil/ plants was treated with the following in these respective quantities:

Ferrous Sulphate	10kg/hq
Gypsum	1000kg/hq
Farm Yard Manure	10tonnes/hq
Sulphur	100kg/hq
Ferrous Sulphate Spray	2%
BIO MAGIC	0.5%

With the exception of BIO MAGIC none of the above applications successfully treated chlorosis. This is evident by the green plots in the front row of the photograph.

RESULTS OF PLOT AND MINI- PLOT TRAILS ON GROUNDNUT WITH BIO MAGIC SPRAY (CY 2005-2006)

		PLOTS				MINI-PLOTS			
TRAIT OBSERVED	MATURITY	CONTROL	TREATED	% INCREASE	% DECREASE	CONTROL	TREATED	% INCREASE	% DECREASE
No of pods	Mature	4.9	7.9	53	-	12.5	20.8	66	-
	Immature	6.1	4.1	-	32	10.8	4.71	-	55
No of Kernels	Mature	7.18	9.0	25	-	16.8	35.2	109	--
	Immature	8.36	7.27	-	13	12.0	8.0	-	33
Wt of pods(g)	Mature	3.8	5.1	34	-	13.39	24.13	80	-
	Immature	2.6	0.8	-	69	2.93	1.32	-	54
Wt of Kernels	Mature	2.6	2.7	8	-	7.81	7.81	111	-
	Immature	1.9	1.5	-	21	2.02	2.02	-	29

Above figures are per plant

Conclusion: The results of the above replicated trials indicate that there is a significant increase in all the desirable parameters essential for increase in yield of groundnut. There is a net increase in number and weight of mature pods with decrease in number and weight immature pods with BIO MAGIC treatment. This could be due to supply of photosynthate required during all the stages of plant growth resulting in net increase yield – the sum total effect of increased chlorophyll in treated and plots.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMERS' FIELDS AT 22 LOCATIONS IN JUNAGADH DISTRICT, GUJARAT , INDIA ,IN 2004 SHOWING EFFECT OF BIO MAGIC ON GROUNDNUT CONDUCTED UNDER THE AGA KHAN RURAL SUPPORT PROGRAMME

NO	NAME OF VILLAGE	NO.OF LOCATIONS	AVERAGE YIELD(kg/Acre) CONTROL	TREATED
1	Hussainabad	4	750	825
2	Samadhiyalaya	4	550	600
3	Hirenvel	4	500	550
4	Sukhpur	5	500	570
5	Chanduvav	5	500	575
			2800	3120
			560	624

Yield of dry pods of groundnut in Kg/acre

As per of programme organizer, the Aga Khan Rural Support Programme there was **50% recovery from CHLOROSIS after first spray and more than 90% after second and third sprays.** Despite erratic/ untimely rainfall during the growing season, there was a **net increase of 11.4% in the yield** of groundnut pods.

RESULTS OF EXPERIMENSTS CONDUCTED ON FARMER'S FIELDS OF BHAVNAGAR DISTRICT DURING THE SUMMER OF 1991 BY T & V STATE DEPARTMENT OF AGRICULTURE. - GUJARAT STATE

NO	FARMER'S NAME	VILLAGE	TALUKA	NO & DATE OF SPRAYS (AKTI M3) (1991)	YIELD IN PODS (KG/HA) TREATED	CONTROL	REMARKS OF DEPUTY DIRECTOR OF AGRICULTURE
1	Madhu D.	Tarad	Mahuwa	Three 29/3, 8/4, 29/4	2430	2165	Very Good results in Yield
2	Arvind. N.	Savarkundla	Savarkundla	One 30/3	1735	1725	No Advantage
3	Nanu V	Savarkundla	Savarkundla	One 31/3	1470	1470	No Advantage
4	Navnit. V	Kundheli	Talaja	Two 30/3, 9/4	2630	2300	Good Results
5	Shamu T	Kundheli	Talaja	Two 31/3, 10/4	2565	2142	Good Results
6	Devji L.	Kundheli	Talaja	Two/ 31/3, 10/4	2700	2435	Good Results
7	Premji P.	Kundheli	Talaja	Two/ 29/3, 8/4	1980	1640	Good Results
8	Damji V.	Kundheli	Talaja	Two/ 31/3, 9/4	2190	1910	Good Results
	AVERAGE				2212	1973	Percentage increase 12% *

- If we disregard the data of the farmers who have sprayed only once then the percentage increase in yield is 15%.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMER'S FIELDS UNDER THE OILSEED PRODUCTION PROGRAMME IN KHARIF,1991.GROUNDNUT CROP GROWN IN PROBLEM AREAS OF RAJKOT DISTRICT, GUJRAT STATE TREATED WITH AKTI-M-3 , 0.5% SOLUTION BY FOLIAR APPLICATION

NO	TALUKA (SUBDISTRICT)	NO OF TRAILS	GROUNDNUT PODYIELD* TREATED	CONTROL	INCREASE IN YIELD(%)
1	Kandorna	4	170	155	9.68
2	Dhoraji	4	580	515	12.62
3	Kothda Sangani	4	425	402	5.72
4	Upleta	4	510	400	27.50
5	Jetpur	4	660	645	2.33
6	Gondal	5	1170	1010	15.84
	Total	25			

*Yield inn Kg/Plot , The results are drawn from the report of Extension officer, T& V Jetpur division. The plot sizes varied. All the trails conducted in one Taluka were considered one unit and percentage increase or decrease in yield as a parameter for comparison. Despite scarcity of rain in Kandoena, Kota Sangani and Jetpur talukas, and considering that in some instances the crop was harvested earlier (due to failure of monsoon rains) the above data clearly indicate a net 12.30% increase in yield even at district level in 25 trails at diverse locations. At locations where protective irrigation was possible, the results are very encouraging.

THE STUDY OF COMPARATIVE PERFORMANCE OF DIFFERENT GROWTH STIMULANTS ON SUMMER GROUNDNUT

NO	TREATMENT			YIELD DRY PODS (KG/HA)				
	STIMULANT	DILUTION	APPLICATION	1990*	1991**	1992***	POLLED	SHELLING%
1	NAA	25 PPM	30 DAS	2933BC	2828AB	1742BC	2501CD	62.8
2	NAA	25 PPM	30 & 40 DAS	3048 AB	2684BC	2138 B	2624 BC	64.4
3	GA	25 PPM	30 DAS	3007ABC	2836AB	1820BC	2554C	64.1
4	GA	25 PPM	30 & 40 DAS	2970ABC	2821AB	1959BC	2584BD	64.4
5	Boric Acid	0.2%	30 DAS	3244 A	2960AB	2072B	2759B	67.7
6	Boric Acid	0.2%	30& 40 DAS	3089AB	2799AB	1542C	2476CD	62.7
7	Photosynthegin	20 PPM	30 DAS	3113AB	2836AB	1939BC	2629BC	61.1
8	Photosynthegin	20 PPM	30 & 40 DAS	2983ABC	2769AB	1725BC	2492CD	63.1
9	BIO MAGIC	0.5%	30,45&60 DAS	3175AB	3031A	2784A	2996A	67.1
10	Control			2765C	2435C	1808BC	2336D	65.5
	S.E.+/-			87.68	92	144	60.6	
	C.D.+/-			248	260	406	194.0	
				10	11.4	22	5	

Data courtesy of Mr.J.J.Macwan, Ph.D. Thesis (Agronomy) ** Data from Agronomy Dept , Gujarat Agricultural University , Anand, India. Interpretation consistent performance of BIO MAGIC merits recommendation for farmers,

APPENDIX IV A : RESULTS OF TRAILS CONDUCTED ON GROUNDNUT CROPS AT VERAVAL IN 1991 UNDER THE AGA KHAN RURAL SUPPORT ROGRAMME (TALUKA MANGROI, DISTRICT :JUNDAGADH)

NO	NAME Of FARMER	VILLAGE	YIELD(Kg/Acre) CONTROL	TREATED	INCREASE IN YIELD(%)
1	Bhana Rama	Karamadi	1000	1135	13.5
2	Bikha Devanand	Karamadi	760	1000	31.5
3	Rama Naran	Kankasa	Vitiated*		
4	Bhura Ramanlal	Kankasa	No harvest at all due to drought		

5	Naran Natha	Kankasa	135	165	22.2

Experiment was vitiated as farmer applied concentrated ferrous sulphate solution, one week after application of BIO MAGIC spray . Average increase in yield was 21.5%

APPENDIX IV B , RESULTS OF TRAILS CONDUCTED ON GROUNDNUT CROPS AT ERAVAL, IN 1992 UNDER THE AGA KHAN RURAL SUPPORT PROGRAMME (TALUKA: MANGROI, DISTRICT: JUNAGADH)

NO	NAME OF FARMER	VILLAGE	YIELD (KG/ACRE) CONTROL	TREATED	INCREASE IN YIELD(%)
1	Bhura Ranmal	Karmadi	480	600	25.0
2	Raju Karsan	Karmadi	600	720	20.0
3	Kana Punja	Karmadi	600	690	15.0

Out of six, three experiments were vitiated due to continuous water logging.Average increase in yield was 20.0%.

RESULTS OF EXPERIMENTS CONDUCTED ON FARMER'S FIELD'S IN KHEDA DISTRICT, GUJARAT , INDIA, 1992. FOLIAR SPRAY OF BIO MAGIC ON SUMMER GROUNDNUT (AS PER RECOMMENDATION OF STATE LEVEL COMMITTEE)

NO	TALUKA	NO.OF TRAILS	GROUNDNUT PODYIELD* TREATED	CONTROL	INCRESE IN YIELD (%)
1	Balasinor	3	1623	1303	24.6
2	Khambhat	6	2162	1996	08.3
3	Anand	4	3575	2865	24.8
4	Thasra	1	2160	2400	-11.1
	Weighted mean	14	2336	2031	15.0

*Groundnut pod yield in kg/Ha ; Conclusion: Results of BIO MAGIC spray on summer groundnut in Kheda district indicate overall 15% increase in yield even after considering negative result obtained at one place because of poor management (about one half of the treated crop was eaten by cattle!) as per report from the Training & Visiting (Agricultural) officer.

APPENDIX VI : COMPARATIVE PERFORMANCE OF BIO MAGIC WITH CONVENTIONAL TREATMENT OF FERROUS SULPHATE+ CITRIC ACID AT GUJARAT AGRICULTURAL UNIVERSITY (ANAND CAMPUS), INDIA. CROP: SUMMER GROUNDNUT

NO	TREATMENT	DRY POD YIELD (Kg/Ha)	SHELLING (%)
1	BIO MAGIC Soil Application at 5 liters/Ha	1239	63.3
2	BIO MAGIC Foliar spray (3 sprays at 30,45 & 60 DAS)	1580	66.2
3	Ferrous Sulphate Soil Application at 50 kg/Ha	1110	63.5
4	Ferrous Sulphate + Citric Acid at 10+2 Kg/Ha (3 applications at 30,45 & 60 DAS)	1077	63.6
5	Nitrogen at 25 Kg/Ha	1361	66.1
6	Absolute Control	1041	64.0
	S.E.m		
	C.D.5%		
	C.V.%		

CONCLUSION: Foliar spray of BIO MAGIC recorded highest pod yield and was significantly superior to application of ferrous sulphate + citric acid. Soil application of BIO MAGIC and ferrous sulphate did not produce significant results.

APPENDIX VII
June 29th , 1996.

WALLACE LABS
TISSUE REPORT
165 Coral Circle,
EI Segundo, CA 90245
(310) 615-0116

TISSUE REPORT
Location Portland, OR
Requester: Kurt Anderson

ELEMENTS	**96-180T-1 shoots**	**96-180T-2 179T-Roots**	**Average Lettuce**
Phosphorus	7,472.47	13,019.88	8,000
Potassium	65,212.08	71,584.79	80,000
Iron	148.80	271.02	75
Mamganese	226.03	106.55	50
Zinc	152.72	136.08	40
Copper	13.36	14.52	10
Boron	36.77	19.72	40
Calcium	16,627.89	8,940.40	30,000
Megnesium	2,553.56	1,955.58	7,000
Sodium	683.15	1,074.42	
Sulfar	2,910.20	2,066.66	
Molybdenum	0.12	0.21	
Aluminium	109.25	253.05	
Arsenic	< 0.13	0.33	
Barium	53.97	69.03	
Cadmium	1.47	0.84	
Chromium	0.40	0.50	
Cobalt	<0.05	<0.05	
Lead	1.81	5.21	
Lithium	1.77	14.30	
Mercury	<0.03	<0.03	
Nickel	4.50	4.18	
Selenium	0.49	0.49	
Silicon	120.34	99.78	
Silver	<0.02	<0.02	
Strontium,	87.88	81.56	
Tin	<0.28	<0.28	
Titanium	4.31	9.79	
Vanadium	0.29	0.81	
Nitrogen			
Chloride			

Units are mg/kg (parts per million) on a dry weight basis expect as noted.
Leaves and roots were rinsed with deionised water

APPENDIX VII WALLACE LABS, 165 Coral Cirlce,El Segundo, CA 90245

ELEMENTS	96-14T-1	96-14T-2	RADISH – STANDARD VALUES			UPTAKE BY SHOOTS (in mg)		
	TREATED	UNTREATED	LOW	SUFFICIENT	HIGH	TREATED Shoots	UNTREATED Shoots	% INCREASE (by treatmt)
Phosphorus	3,174.0	3682.32	2,500-2,900	3,000-7,000	>7,000	3.6627	0.8703	421%
Potassium	38,922.2	28397.93	35,000 39,900	40,000-75,000	>75,000	36.7918	6.7114	548%
Iron	598.2	514.07	40-49	50-200	>200	0.5669	0.1279	443%
Manganese	44.5	38.80	35-49	50-100	>100	0.0422	0.3092	460%
Zinc	227.1	174.08	15-18	19-100	>100	0.2150	0.0411	522%
Copper	16.0	12.37	3-4	5-25	>25	0.0155	0.0029	530%
Boron	25.8	26.04	20-24	25-125	>125	0.0245	0.0062	398%
Calcium	22,589.0	21,473.74	20,000 29,900	30,000-45,000	>45,000	21.3523	5.0750	421%
Magnesium	3,742.7	3,348.01	3,000-4,900	5,000-12,000	>12,000	3.5372	0.9912	447%
Sodium	1,462.6	736.79				1.3829	0.1741	794%
Sulfur	5,999.4	4,650.68				5.6708	1.0991	516%
Molybdenum	4.7	7.18				0.0038	0.0017	227%
Aluminium	1,058.0	943.16				1.0005	0.1229	449%
Arsenic	0.6	0.83				0.0008	0.0002	413%
Barium	59.9	73.19				0.0561	0.0173	325%
Cadmium	0.4	0.15				0.0004	0.0000	1186%
Choromium	1.4	1.13				0.0012	0.0003	440%
Cobalt	<0.5	<0.05						
Lead	7.7	6.69				0.0069	0.0016	435%
Lithium	1.3	1.12				0.0012	0.0003	441%
Mercury	<0.3	<0.03						
Nickel	2.6	1.43				0.0020	0.0003	578%
Selenium	<0.9	0.49						
Silicon	391.0	374.23				0.3699	0.0884	418%
Silver	0.3	<0.02				0.0002		
Strontium	124.8	135.98				0.1178	0.0321	366%
Tin	0.8	<0.28						
Titanium	20.5	20.75				0.0196	0.0049	400%
Vanadium	2.1	2.00				0.0021	0.0005	444%
Nitrogen	4.9%	6.45%	2.8%-3.0%	3.0%-6.0%	>6.0%	46.7899	15.2435	307%

Units are mg/kg (parts per million) on a dry weight basis expect as noted.
Requester: Kurt Anderson Dry weight per plant in grams. Plant: Radish

Shoots	0.95	0.236
Roots	0	0.17

EFFECTVENSESS OF METHODS DEVELOPED AND EMPLOYED FOR CONTROLLING CHLOROSIS BY USE OF BIO MAGIC.

INTRODUCTION

In chlorosis affected leaves thylacoids- organelle where chloroplasts manufactured – are badly affected and do not produce adequate chlorophyll. Grana sacks are greatly reduced (from the desired 12 to 14 to a mere 3 to 7) rendering chloroplasts inefficient in utilizing solar energy thereby affecting the process of photosynthesis. For managing such situations frequent spraying of micronutrients –containing formulations are widely used but this practice has certain demerits. Another approach is to apply soil amendments, mainly use of sulphur containing fertilizer, copious supply of FYM, acidification of soil etc. Despite this the desired results are not always obtained. In advanced countries pressure injections in tree trunks are practiced but these are expensive, require skilled technicians and hence are cost – prohibitive. We therefore tired to develop methods which are simple, easy to practice and cost effective. And after prolonged experimentation, research –both in the laboratory and in the field , continuous monitoring , assessment and review , we are confident that in BIO MAGIC we have a product that addresses the problem, shows results within 72 hours, is simple to use and is cost- effective. In addition, being of origin, it is non –poisonous and eco – friendly. The following methods of application of BIO MAGIC were all tired.

SPOT APPLICATION

This method id primarily meant for studying effectiveness in absorption, penetration, proliferation and time required for regreening of treated parts. Twings of chloratic trees are selected keeping in view age and degree of chlorosis of trees to be treated. In bifurcated twigs one branch is kept as treated and another as control and continuous monitoring is done to study regreening. In the branch selected for treatment a list is made in the bark and BIO MAGIC was applied with specific carrier material.

DRIP DROP FEEDING

In this method a hole is drilled (depending on diameter of branch to be treated) and BIO MAGIC liquid (diluted 5 times) is dropped into the hole at three drops per minute.(IV sets discarded by hospitals /clinics were used for this purpose as they worked out cheapest.)

ROOT FEEDING

This is useful in shallow rooted trees like citrus , coconut ,fig , cashew nut , grapes, etc. Absorbing root (about 1 cm in diameter) is selected and a cut with a desired angle is made on the underside. BIO MAGIC (duluted 5 times) is fed to absorbing root. This last method ,viz Root Feeding is still under trail and various experiments are in progress. Results communicated to us are highly encouraging ,but photographic records are not available yet. Reports from Nangadon in Maharashtra , India indicate hat coconut trees treated by root feeding have flowered for the first time in ten years. As soon as research and trails completed, photographic evidence to demonstrate the effectiveness of this method will be made available.

EFFICIENCY OF BIO MAGIC- TRIAL REPORTS AND RESPONSES FROM SPECIFIC CROPS TO BIO MAGIC TREATMENT

EFFICIENCY TESTING OF BIO MAGIC ON SAPOTA (CHICU)

Sapota has been reported to suffer from chlorosis. Shorrocks (1984) has included Sapota in the list of trees sensitive to chlorolsis. Trees affected by chlorosis bear fruit of smaller size and / or in lesser number and although leaves may not show the yellow coloring they are of a paler green compared to the leaves of normal trees.

PLATE I

As mentioned earlier, one half bifurcated twig was treated (T) with the other half forming the control (C). This picture, taken 22 days after application of BIO MAGIC, shows the treated leaves darker in colour indicating that the nutrient has flowered whilst the control is yet to flower. Prior to treatment this tree brown few fruit and the fruit were small. It is also worth nothing that the mid-ribs became dark green in colour and then proliferate to secondary, tertiary and other veinlets.

PLATE II

Branches that were spot – treated with BIO MAGIC are dark green in colour when compared with the pale green of the control branches (upper branches). Treated branches show a perceptible increase in chlorophyll. The size of leaves has also increased and flowers profuse. This effect was observed six to eight weeks after treatment. In winter when movement of plant sap is slower perceptible effect is visible after about twelve weeks.

PLATE III

Branches spot-treated on different dates. Note the darker green and healthy appearance of foliage on treated branches. The difference between different treatment dates is clearly visable.

PLATE IV

This plate shows the results of Spot Application (SA). Bark of branch has been removed by making two parallel incisions 7cm long and 5 cm apart. One more incision is made at distal end from this end bark is slowly and carefully lifted till the other end (which is intact). BIO MAGIC paste is inserted and the bark replaced and the incision covered with a plastic tap shown. Direction of branch is shown by hand of researcher. Treated branch has dark green loves. Othet tewo branches near SA are pale green. Treated branch also flowering.

PLATE V

This plate shows how greening starts after treatment. The leaves marked "G" first from top have greened completely, second "G" also shops complete greening , whereas in third :G" greening is not yet complete. In the leaves marked "PP" regreening in striations along the mid – rib of leaf can be observed. In other leaves of treated branches profileration patterns (PP) are visible. This shows the path of nutrient supply plays an important role in regreening after treatment.

PLATE VI

Treated and control branches are shown in close proximity. Treated (SA) branches show much earlier flowering , increased size of leaves , dark green and shining foliage. Control is pale green with no or negligible flowering. A portion of foliar sprayed branch can be seen at extreme right showing green spots but greenness is unlike that SA treated branches.

PLATE VII

This branch was treated on 24th November, 1995, i.e. wit the onset of winter (November to February) to identify best period of treatment (SA) in evergreen tropical fruit tree (Sapota). Details such as chlorophyll appreciation, onset of flowering etc. were obsereved and recorded together with photographs taken periodically. This picture was taken 5 months after treatment (in April,1996). Note that treated branches are dark green , have profuse flowers, and increased size of leaves. This also indicates that SA has longer lasting effect.

PLATE VIII

This plate shows branch of Sapota tree located in University area about 10 km from Anand. Tree was chloratic and locate near compound wall. The tree was to be cut down for the expansion programme and was ideal for our experiments. We requested the authorities to retain the tree till we completed our trails. Treatment was carried out on 25th January ,1995, (just warming into winter), with SA. Note the severity of chlorois. Treatment was applied on a branch having about 10cm diameter. Slant incisions were made on the branch and BIO MAGIC paste was inserted into the incisions which were then covered with plastic tape. Picture shows effect of SA. Marked difference is noticeable between control. (C) and treated (T).

PLATE IX

Another picture of the referred to in plate 8. Effect of SA has persisted and the treated branch shows dark green leaves and healthy fruit. Flowers that have appeared on control branch are weak, shriveled and non- fruit bearing. Leaves of control branch are highly chlorosis and smaller in size.

PLATE X

Another view of the tree shown in plates 8 & 9. difference between treated and control is spectacular.

PLATE XI

A picture of the Sapota tree referred to in Plate VIII(p 9). Untreated portion of this tree became progressively chloratic as can be seen in the picture taken on 3.11.95 (Plate VIII). Brach in the centre of the picture having dark green colour was treated by Spot Application (by slit opening and filling with BIO MAGIC paste). Picture clearly shows and demonstrates effectiveness of BIO MAGIC treatment.

PLATE XII

This picture shows the results of Drip Drop method of applying AKTIM M-3 : a hole was drilled into the branch near while label (partially visible at centre of picture). 10 ml of BIO MAGIC with 50ml of water was filled in IV sets (as used in saline drip in hospitals) and drip dropped into drilled hole at 3 drops min for 24 hours. Part of solutions fell on the ground. Treatment was given on 26 November , 1995. Results were spectacular as can be seen in the picture above. Foliage of the treated branch attained dark green colour with shining leaves. Control (seen at right of picture) is pale green and dull as against dark and shining leaves of treated branch.

PLATE XIII

One of the techniques of in the management of cholorosis is foliar spray and is being used widely. It has the demerit of requiring frequent applications. It also imparts spottiness to fruit and hence limits its use. In this picture of spray application of BIO MAGIC, dark green spots appear on leaves and degree of effectiveness is not to the level of that obtained by Spot Application or Drip Drop treatment as we have seen earlier.

EFFICIENCY TESTING OF BIO MAGIC ON MANGO (MANGIFERA INDICA)

INTRODUCTION:

Mango (Mangifera Indica) has been reported to be affected with chlorosis especially in orchards raised on carcareous soils. Mango is also a tropical evergreen tree and 75% of fruit are produced in India. Chlorosis is often associated with die-back disease having unknown etiology. Recently :Bandi" (local name of disease in "Kesar" growing area of Gujarat state , India) is reported on the increase. In this disease young leaves at the terminal portion of branches become chloratic and malformed and sometimes fall off giving cactus – like appearance to terminal branches. The tip of the effected branch becomes necrotic and eventually dies. New buds do sprout from lower side of affected branch but usually meet the same fate. The Kesar variety of mango has a delicious taste and pulp of golden colour. It also has specifically appealing flavour, sweetness and keeping quality and has earned credit and is great demand from both Indian and export markets. A survey of the area showed that more than 50% of the trees were affected with " Bandi". Not all trees in every orchard were affected. Trees not affected were healthy and showed normal yield performance. "Bandi" also strikes in young orchards and sometimes all the saplings in an orchard have to be uprooted and new ones planted- causing a loss of time and money.

At village Rampol, about 25km from Anand, a farmer who had an orchard of 216 Kesar mango trees decided to destroy the orchard because all the saplings (about 3 to 4 years old) showed die – back symptoms. We requested him not to uproot the saplings started to sprout new buds with dark green leaves and subsequently all the saplings were treated with SA. Now he has decided not to uproot the saplings.
Similarity, at village Talala of Junagadh District which is the heartland of Kesar mango, "Bandi" was observed at several orchards. Following the successful treatment at Ramol we also undertook to treat the trees at Talala. Later we had reports of "Bandi" in other areas in Gujarat and all were treated on a trail basis (SA, DD and RF). Results and feedback coming from various locations are encouraging. There are indications that such kind of treatment may prove helpful in chlorosis induced die- back and restoring orchards.

PLATE XIV

This picture (taken at village Ramol) shows symptoms of die-back on Kesar mango trees. Younger leaves are chloratic to begin with , they then fall off gradually and the terminal portion becomes dead and appears coctus –like. Lower living porlion sprouts leaves which eventually die and fall off. In necrotic tissue secondary infection takes place especially saprophytic fungi like Botryodiplodia spp , a rot – causing fungus, associated with other saprophtes. Prescribed treatment against this does not sustain. This was what was happening on almost all the plants at Ramol.

PLATE XV

As the trees were young, (3 to 4 years) SA treatment was applied on 14.8.95. (during the raily season). New buds started to sprout after 9 days (as seen in this picture). Note the remenant of dead nacrotic leaves. Many new buds can also be seen.

PLATE XVI

SA portion of mango graft is shown in this picture. Note the dark patches on stem just below label and also on right branch. Cell sap movement in conducting system of plants being rainy season results appear fast.

PLATE XVII

This picture shows mango graft at Ramol treated on 21.9.95. Photo taken on 3.10.95 (12 days after treatment) shows new buds. All treated plants are monitored every month and almost all have been restored to full health. After seeing the results of treatment farmer has decided to retain all the mango saplings(and not to destroy them as earlier intended).

PLATE XVIII

This picture (taken on 3.10.95) shows one of the plants treated on 14.8.95. Note the greening of the leaves. The green patches are seen patches are seen on the top and the lowest leaves. The top branch is formed after the treatment, new buds that have sprouted have now become branches. Marginal necrosis is no more to be seen.

PLATE XIX

Another picture (also taken on 3.10.95 of plants treated on 14.8.95) showing dark green shining leaves of larger size. Note the new shoots having anthocynin pigmentation. New branches healthy appearance are seen on the left. When chlorosis is present new leaves take longer time to attain normal size whilst in treated plants leaves attain normal size in shorter time. Overall appearance of healthy and of a good complexion.

PLATE XX

This picture shows a view general view of the mango orchard at Ramol village mentioned earlier. Remmants of chloratic plats are still visible in this field. All the mango plants were treated on 14.8.95.

PLATE XXI

Owner of mango orchard at Ramol, Mr. Rajnibhai patel, is closely observing the effect of treatment and showing dead leaves of the plant. Completely new branch with dark green leaves can be seen.

PLATE XXII

Visitors from overseas at the SA treated orchard at Ramol to observe performance of BIO MAGIC on sapota , groundnut mango ,eucalyptus, vegetables etc. other countries also having problem of chlorosis in many sensitive crops.

PLATE XXIII

This plant is located at Nadiad (25 km from Anand) and it is a classic example of chlorois followed by die-back symptoms. Prior to treatment all the leaves had marginal necrosis and were dull in appearance. Remenants of die-back can be seen in this plant on second branch form the top. Top branch shows new shoots grown after SA treatment. Two new shoots were observed 41 days after treatment – one terminal branch and another bottom branch. Note the shining green colour of the new leaves.

PLATE XXIV

This picture shows the top branch of the plant shown in Plate X. The three leaves at the left have dark green colour whilst the fourth leaf at the bottom centre of the picture is still in the process of greening. This picture was taken 41 days after SA.

PLATE XXV

After the success of SA and DD treatments at Ramol farmers from Ahmedabad approached us and on 9.12.95 a team of technicians was deputed to assess the magnitude of the problem. Survey indicated that the problem is quite serious. This picture shows an 8- year old plant. Chlorosis marginal necrosis can be observed. Note that the growth of the tree is not commensurate with age of the tree. After seeing the report we decided to give SA treatment to smaller plants and DD to larger trees.

PLATE XXVI

This is a close-up view of tree to be treated at village Surva in Talala (450 km from Anand). SA treated branches are indicated by arrows. Die –back symptoms are clearly visible. Even newer branches (top right) have leaves with mariginal necrosis.

PLATE XXVII

This picture shows an affected tree from a mango orchard situated at village Borav of Talala taluka and the condition of the orchard is almost the same as that observed at Surva. Because of increasing demand for :Kesar" mangoes acreage under this variety is increasing and the problem of die-back is also on the increase. We have treated many orchards on a trail bais. Preliminary results are very encouraging. All trees treated being constantly monitored.

EFFICIENCY TESTING OF BIO MAGIC ON GROUNDNUT (PEANUT) (ARACHIS HYPOGAEA)

INTRODUCTION

This is the general layout of the experiment. Soils weighing about 4 MT from each of the areas known to have a history of cholorosis were brought. Each load (of 4 MT) was thoroughly mixed and then dividend into two equal portions and placed in mini-plots (2m X m X 1m) soils brought were from Amreli, Junagadh (Navivadi), Panchmahals, & Junagadh seed farm.

Amereli

In this district about 400,000 ha of land is under ground-nut during June- July and October- November, i.e. during the onset of the rainy season. Out of this about 40%-50% of the area suffer from mild to chronic chlorosis –mainly Ca –induced iron chlorosis. Soils have 16-70% CaCO3, are low in (nitrogen) cotent, poor in available P () but have adequate K( ) some of the soil samples show adequate Fe but still suffer from chlorosis. This may be due to high bicarbonate content interfering with uptake by plants.

Junagadh (Navivadi)

These soils are known for iron deficiency and have a history of chlorosis since 1996. They are under the charge of Agril Reasearch Station. Soil analysis CaCo3 -18% N 0.01% P available 9-15 Kg/ha. K-364 Kg/hs. Peanuts grown in any season start showing chlorosis within 10 days after germination.

Panchmahals(Derol Seed Farm)

I this district mainly maze and upland paddy are grown during rainy season. Upland paddy covers approximately 80,000 to 100,000 ha depending on onset of rainy seson. Paddy from mild to severe chlorosis resulting in complete failure of crops in some of the areas. Soils contain CaCo3 -11% ,N-0.09% p (available) 15kg/ha.

Junagadh Seed Farm

As in Navivadi this area has also known history of chlorosis . Soil content: CaCO3 -38% N -0.085% P (available) 15kg/ha.

None of the plots were supplied with F.Y.M or compost. Groundnut cv GG2 was sown in the month August,1995. 50 seeds were planted in a row, with rows 30cm apart. The groundnut plants were treated with BIO MAGIC (5ml/L) 30 DAS (first application), 45 DAS (second application) and again 60 to 65 DAS (third application.).

PLATE XXVIII

This picture was taken after the first application of BIO MAGIC. Front plots are f Amreli soils (left plot is untreated , right treated), second row of plots is Jundagadh soils (left treated , right untreated) , third row Panchmahal soils (left untreated right treatd) and fourth row (not visible in the picture) Junagadh (Seed Farm) soils (left treated, right untreated). Results clearly indicate the efficacy of AKTI M- as can be seen from the picture.

PLATE XXIX

A closer view of the plots containing Amreli soils taken after the first treatment. Stunted growth of interveinal cholorosis is clearly visible in the untreated plot. It can

also be noted that some of the plants have succumbed to the disease- a few gaps are visible (the left row for instance.). the shows plants with increased height and girth and having dark green foliage. This shows increase in chlorophyll content. Final survival rate was 92.7% (139 plants) for the treated plot compared with 56.7% (85 plants) for the untreated plot.

PLATE XXX

The two plats shown in this picture contain Junagadh solid. These are a better indication of the effectiveness of remedial treatment because of the known history of chlorosis. After just one treatment the difference between the plants in the treated plot and the ines in the untreated plot is obivious. Note severe chlorosis and stunted growth in the untreated plot.

PLATE XXXI

The soils in these plots are from Junagadh seed farm. Severity of chlorosis is less compared to soils from Amreli (Plate XVI) and Navavadi(plate XVII). How ever ,results of treatment with BIO MAGIC are very clear. In the untreated (control) plot some of th plants show irreversible chlorosis. Final survival rate was 81.3% (122 plants) for the treated plot and 74.7% (112 plants) for the untreated plot. Under good climatic conditions productivity of these soils is very good. The (Junagadh) district has 600,000 hectares under groundnut in the rainy season. Chlorosis is mild to severe. In soil survey about 40% of soil samples were found to be deficient in Fe and Zn.

PLATE XXXII

This mini-plot is yet another from Amreli soils. Treatment effects are visible on chloratic leaves as dark green spots and patches. Green leaves also show more greener areas suggesting effectiveness of BIO MAGIC. Even leaves which are very chloratic now have about half their surfaces covered with green patches (see rear of row on right). Treated plants also flowered earlier.

PLATE XXXIII

This picture of the plots containing Amreli soils was taken on 28.8.91 after complection of three applications of BIO MAGIC. The treated plot (right) shows profuse growth with rich, dark green foliage. From hese plots 14 plants were selected at random from each mini-plot for study of effect of treatment on various parameters of groundnut plants. Results were presented on national seminar on Groundnut Production Technology. There desirable parameters for groundnut production.
Summary of results are as follows:

Number of pods	66% increase
Number of kernels:	109% increase
Weight of pods:	80% increase
Weight of kernels	111% increase

PLATE XXXIV

This picture shows the field trails at Navivadi Farm, Junagadh. This trial was conducted after the success of the mini- plot trails. The purpose was to compare the efficacy of BIO MAGIC with other treatments. This location was chosen because it

has a known history of chlorosis. Three year pooled results showed the efficacy of FeSo4+ Citric acid and BIO MAGIC to be at par, but when input cost benefit ratio was considered BIO MAGIC worked out cheaper.

For details of results please refer to Appendix III (Book1, Page27)

PLATES XXXV to XLIII:

In these plates effort has been made to show the effects of foliar spray especially period of absorption, time required to start regreening , physiological stage of leaves etc. The technical literature reports showed that Fe gets absorbed at the site of application within 24 hours.

PLATE XXXV

BIO MAGIC was sprayed on 3.4.91 and appearance of regreening symptoms was monitored every six hours on the second leaf from terminal bud of this groundnut cv GG2 30 DAS. Spraying was done at 1800 hrs (i.e. late in the evening) at concerntration of 5ml BIO MAGIC to I litter of tap water. Groundnut seeds were planted in pots containing solid from areas with known history of chlorosis. Intervenal chlorosis appeared one week after germination. Results of application: second leaf from terminal bud showed regreening within 24 hours. In detailed observations, regreening started just after 12 hours (early in the morning at 0600hrs). interval chlorosis is clearly seen.

PLATE XXXVI

This picture shows effect of spraying on second leaf from terminal bud 3 days after spraying (refer pate XXXV). Regreenigng has also started on other leaves.

PLATE XXXVII & XXXVIII

These two plates show condition of plants cv GG2 and CV Florida Runner in control plots. Note the severe chlorosis (intervenal chlorosis) in young leaves. This will lead to marginal necvrosis and die-back (refer plates XXXIX and XL).

PLATES XXXIX & XL

These two plates show condition of plants in plates XXXVII and XXXVIII after the intervenal chlorosis had progressed to marginal necrosis and die-back.

PLATE XLI

Groundnut plant treated with BIO MAGIC foliar spray on 19.8.95. This picture was taken 9 days later on 28.8.95. Note the gradual disappearance of chlorosis (refer Plates XXXVI & XXXVII). Intervenal chlorosis can be observed on uncovered areas of sprayed leave. This proves that through spraying is essential for complete results.

PLATE XLII

This picture shows effect of BIO MAGIC on groundnut cv GG2 commonly grown in Gujarat growing area. Proliferation of spray can be noted in the picture. Note also the spreading patterns depending on physiological status of different leaves.

PLATE XLIII

Groundnut cv Florida Runner is observed to be highly sensitive to chlorosis in Gujarat and was taken up for study in 1995. The above picture shows spectacularly the effect of BIO MAGIC on the plant (refer plate XXXiX for control).

PLATES XLX, XLVI, &XL VII

These plates show a specific technique (the half –leaf test) developed to appreciate chlorophyll contents periodically as affected by spraying of BIO MAGIC at recommended dose.

PLATE XLV

Leaflets of the same compound leaf of the same plant exhibiting typical symptoms of intervenial chlorosis are selected and half of the leaf is brushed with BIO MAGIC (using a swab of cotton) whilst the other half is retained as control.Greening in the treated portion is clearly seen in this picture whereas in the control portion clorosis is appaent. This technique nullified , or very nearly nullifies the problem of defferential response of physiological status of leaves in various stages of growth. Periodic chlorophyll content was estimated by Hiscox and isralestan (1979) method.

PLATE XLVI

This plate demonstrates the efficiency o the technique mentioned above. Upper –hald , lateral –half treated plants are shown with clarity in this plate. Severely chlratic leaves with marginal necrosis are also visible.

PLATE XLVII

Graph shows periodic chlorophyll appreciation (green –BIO MAGIC. yellow –control ,red commercial product). Chlorophyll content was almost the same at the beginning of the experiment . However whilst the plant treated with the commercial product and the control (untreated) plant showed decrease of chlorophyll , the plant treated showed increase of chlorophyll. Peak in chlorophyll was obtained on the 10th day after treatment. Thus it s advisable to apply second treatment between 10 to 15 days of first application.

PLATES XLVII TO LVI

A few more experiments were conducted to test the improved version (BIO MAGIC) during the 1995 rainy season. Results showed regreening of treated plants was spectacular and this just after one spry! The following pictures illustrate this.

PLATE XLVIII

This two mini-plots (96 &97) having soils from the same piece of land. Cv Rlorida Runner was grown during the rainy season of 1995 (sowing was done on 20.7.95) and both plots showed the same degree oc chlorosis. Plot 96 was then subject to two spraying of BIO MAGIC (once on 19.8.95 and again on 28.8.95. The treated plot shows dark green colour of foliage with little of no evidence chlorosis,

PLATEXLIX

The control plot (pot 97) shows many chloratic plants with stunted growth. The efficacy of BIO MAGIC in treating chlorosis is evident.

PLATES L& LI

These two plates show three mini-plots of Florida runner. Plot 86 was retained as control , whilst the other two (85&87) were subject to two sprays of BIO MAGIC (on 19.9.95 and 28.8.95) .When plates are viewed together the profuse growth in plots 85& 87 and the absence of chlorosis is obivious.

PLATE LII

These pictures show side view of five mini-plots of floride Runner. Picture was taken on 28.9.95. just before second spraying of BIO MAGIC on treated plots. The foremost two plots and the fourth are control. The third and fifth are treated. The result of just one application of BIO MAGIC speak for themselves.

PLATE LIII

A front view of the mini- plots shown on the previous plate(Plate LII). First plot from left has been treated , and those on either side of it control. The effect of BIO MAGIC in including increase in chlorophyll content in chlorophyll content in chloratic can be clearly seen.

PLATE LIV

The above picture shows two mini – plots , the one on the left being the treated and the other (right) control. The treated was subject to two applications and was due for a third application between 7-12 September, 1995. However the spray was not done to see if chlorosis would reappear. This picture was taken on 21.9.96 and shows that although slight chlorosis did reappear in the treated plot it was not generally contained, compared to the control plot.

PLATE LV

Florida Runner after two sprays of BIO MAGIC profuse growth is noticeable and regreening of some chloratic plants is visible. Picture was taken 16 days after second spray and upper few leaves (especially the new growth) show chlrosis –hence the necessity for three sprays.

PLATE LVI

The control plot of Florida Runner. Compare this with plate LV to note the prevalence og chlorosis and the stunted appearance of the control plants. This is without BIO MAGIC.

EFFICCACY TESTING BIO MAGIC ON LAGEROSTROMIA SPP

INTRODUCTION

Lagerstroemia is an ornamental plant cultivated for its beautiful foliage and profuse flowering which is from may to Octomeber. The tree which is the subject of our study was located near Karamsad town in the landscaped area of a private dwelling and in our view it was a perfect model for various studies related to chlorosis e.g. penetration , regreening , movement of applied nutrient in plant, direction and chlorophyll appreciation , suitable dare for getting response, etc. The plant was about ten years old and twenty feet tall. The leaves were showing typical symptoms of chlorosis and according to the owner of the property, it had not flowered for the past three to four years. The leaves had turned yellow and despite regular application of necessary manure supplements and fertilizers the plant had not become healthy at all. He was quite willing for us to conduct experiments on the tree. Spot Application of BIO MAGIC was carried out on 18th May, 1995 on twigs, leaves ,branches base of thick branches and trunk (as we gathered sufficient experience in treting Sapota, Mango, Eucalyptus , Terminilia , etc). The following pictures show the resultant changes brought about by application of BIO MAGIC at different stages.

PLATE LVII

This plate shows the extent of chlorosis on the Lagerostromia. Leaves exhibit classical symptoms of inter –venial chlorosis where veins are green but area in between is chloratic (i.e. devoid of chlorophyll or less chloroplasts). Dark brown mark seen on front leave are marks made by spot –application of BIO MAGIC (applied on mid-rib on 18.5.95). Area around point of application becomes necrotic because of injury and slight toxicity of concentrated material. Three leaves were treated although in this picture spot on middle leaf is not visible as it is obscured by top leaf (just visible below centre of botoon edge of top leaf). This photograph was taken on 22.5.95 (four days after treatment) and in the top leaf it can be seen that toxicity has spread in the spread systematically through the veins. In the second leaf this has not happened as the nutrient was placed just near the mid-rib (not a in the first leaf). In the third leaf slight injury can be noted on the secondary vein nearest the point of application. There is no difference between spot – treated leaves and those not treated four days after treatment excepting phytotoxicity in the first leaf. Other control leaves seen in this picture are almost similar to the treated leaves.

PLATE LV III

This photograph was taken on 27.5.95 (nine days after treatment and shows the same ethree leaves (c.f. Plate LVII). In the first leaf toxicity has advanced further an greening is observed just around two secondary ribs. The effect of the treatment is more noticeable on the second leaf which clear green area below. The point of application indicating greening effect of BIO MAGIC has already started . similar though not so oblivious effect can be noticed on the third leaf. Backward movement of nutrient is almost negligible which means (indirectly) that BIO MAGIC has entered in xylem flow of conducting system of plants.

PLATE LIX

This photograph slows the same leaves on 7.6.95. (20 days after treatment). Toxicity has further advanced on first leaf. Second and third leaves show more greening. Control leaves are still having inter – veinal chlorosis. (the greening of leaves on branch at right hand side of picture will be discussed on pages 61 & 62).

PLATE LX

The picture shows the same branch of the same tree. This branch was treated by spot- application on 18.5.95., but this time two terminal portions of the branch were selected for treatment. In the branch at the front of the picture SA was applied at point near label, and on the other branch near hand of researcher. Leaves have become dark green after just nine days. (c.f. Plates LVII- LIX). This could be due to faster rate of sap flow towards terminal portion. Note also the tree treated leaves shown in the earlier plates (top centre of picture). Only slight greening has taken place in these leaves. This may be due to difference in absorption and conductance in various parts of the plant.

PLATE LXI

This picture shows the same branch as on page 61 but photographed on 31.5.95(13 days after treatment). Note that all the treated portion has become darker green. (compared to previous plates). Note also that untreated parts have remained in the same condition as at the beginning of the experiment..

PLATE LXII

This picture shows both the parts of the same branch which were treated on 18.5.96 as they were on 7.6.95. To highlight the effects of treatment another branch has been included. THE EFFECTS OF TREATMENT ARE SPECTACULARLY VISIBLE WITH CONTROL BEING SEVERELY CHLORATIC. Development of chlorophyll is remarkable in treated leaves as compared with pictures taken on 22.5.96. (Plate LVI), 27.5.95. (Plates LVIII & LX) AND 31.5.95 (Plate LXI). Front branch shows new healthy leaves and control branch at extreme has new leaves which are still chloratic. These experiments show conclusively that SA and DD applications of BIO MAGIC are an effective solution for treatment of chlorosis in large trees and saplings.

EFFICACY TESTING OF BIO MAGIC ON PADDY (UPLAND & LOWLAND)

PLATE LXIII

These two plots were filled with soils brought from Derol Seed Farm in the Panchmahal districy of Gujarat State, India, where upland paddy is being grown. About 60,000 Ha are under paddy cultivation in this area. The paddy suffers from mild to severe chlorosis, and at times the attack is devastating. Currently varieties tolerant to chlorosis are being grown, though a few areas continue to grow the conventional varieties (e.g. Sarthi 34-36) because of early maturing trait. Spraying of BIO MAGIC was done on eleventh day after germination (it has been observed that chlorosis sets in between ten to twelve days after germination). This picture was taken 26 days after germination (i.e. 15 days after treatment). The plot on the left is the control whereas the plot on the right has been treated. Mark the effect after just one spray! At time of experiments all the plants in the control died. Survival rate in the treated plot was 76%.

PLATE LXIV

This mini plot was dividend into 24 micro- plots to study efficacy in same –soil conditions. The experiment consisted of 6 replications with four treatments which were:

(1) Ferrous sulphate soil applicaton @ 50kg/ha.
(2) Gypsum @ 1 tonnws/ hq.
(3) FYM@ 10 tonnws/hq
(4) Control (spraying of BIO MAGIC)

The micro-plots were dividend by glazed porcelain tiles. As can be seen from the picture only the micro-plots which received foliar spray survived.

PLAT LXV

The picture shows a close up of the micro-plots described in Plate LXIV. Devastating effects of chlorosis can be noted in plots which were not subject to BIO MAGIC treatment. Some chlorosis is observable in the treated micro-plots also, but the survival rate is very much higher at about 70- 76%.

PLAT LXVI

Yet another close-up of the micro –plots showing the effects of chlorosis on untreated plots. Compare BIO MAGIC treated plot with those subject to other treatments.

PLAT LXVII

The tree rows of pots on the right were treated with BIO MAGIC. compare the results with the control (two centre rows) and those subject to other treatments(three rows on the left).

EFFICACY TESTING OF BIO MAGIC ON OTHER PLANTS

POMEGRANATE

PLATE LXVIII

This tree was located in the backyard of a house and according to the Gardner it never flowered adequately and failed to retain fruits. Different branches were treated by SA in January . 1996 , and gardner informed us that the treated tree not only flowered but also retained its fruit. This picture shows full view of tree after treatment.

PROGRANATE

PLATE LXIX

Another view of the same treated tree this time in close-up to show retained fruit. After treatment all the leave had become dark green and shining.

CAPSICUM

PLATE LXX

A farmer who raises capsicum seedlings on a large scale and sells in bulk sprayed BIO MAGIC twice to make the seedlings more vigorous and observed that the sprayed plots had greater number of transplantable seedlings of Capsicum and were healthier than control. The difference between the sprayed and the control plots can be seen in this picture. (sprayed bed is on the left, and the control bed on the right).

MUNG (VIGNA RAIATA)

PLATE LXXI

Mung is an important pulse crop in India and is grown in poor and light soils throughout the country. It frequently suffers from chlorosis and is attacked by pests and disease resulting in heavy crop losses. BIO MAGIC was tired as a treatment for control of chlorosis. The picture above shows two beds of mung plants – the bed on the right has been treated with BIO MAGICand the bed on the left is the control. No other treatments or soil amendments have been used.

MUNG (VIGNA RAIATA)

PLATE LXXII

This picture also shows two plots of MUNG pants. The left plot was treated with twice BIO MAGIC. The pot on the right was used as control and left untreated. Both plots received all other soil amendments viz. FYM phosphorous , sulphur and recommended dose of fertilizers. Note that the left plot is healthier when compared to the control (right).